

# *Drosophila* HCN mediates gustatory homeostasis by preserving sensillar transepithelial potential in sweet environments

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## Abstract

Establishing transepithelial ion disparities is crucial for sensory functions in animals. In insect sensory organs called sensilla, a transepithelial potential, known as the sensillum potential (SP), arises through active ion transport across accessory cells, sensitizing receptor neurons such as mechanoreceptors and chemoreceptors. Because multiple receptor neurons are often co-housed in a sensillum and share SP, niche-prevalent overstimulation of single sensory neurons can compromise neighboring receptors by depleting SP. However, how such potential depletion is prevented to maintain sensory homeostasis remains unknown. Here, we find that the *Ih*-encoded hyperpolarization-activated cyclic nucleotide gated (HCN) channel bolsters the activity of bitter-sensing gustatory receptor neurons (bGRNs), albeit acting in sweet-sensing GRNs (sGRNs). For this task, HCN maintains SP despite prolonged sGRN stimulation induced by the diet mimicking their sweet feeding niche, such as overripe fruit. We present evidence that *Ih*-dependent demarcation of sGRN excitability is implemented to throttle SP consumption, which may have facilitated adaptation to a sweetness-dominated environment. Thus, HCN expressed in sGRNs serves as a key component of a simple yet versatile peripheral coding that regulates bitterness for optimal food intake in two contrasting ways: sweet-resilient preservation of bitter aversion and the previously reported sweet-dependent suppression of bitter taste.

### eLife assessment

This study provides **important** new insight into how non-synaptic interactions affect the activity of adjacent gustatory neurons housed within the same sensillum. The conclusions are supported by **convincing** electrophysiological, behavioral, and genetic data. This work will be of interest to neuroscientists studying chemosensory processing or regulation of neuronal excitability.

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## Introduction

Glia-like support cells exhibit close physical association with sensory receptor neurons, and conduct active transcellular ion transport, which is important for the operation of sensory systems<sup>1</sup>. In mammals, retinal pigment epithelial (RPE) cells have a polarized distribution of ion channels and transporters. They provide an ionic environment in the extracellular space apposing photoreceptors to aid their light sensing<sup>2</sup>. Likewise, knockdown of *Drosophila* genes encoding the Na<sup>+</sup>/K<sup>+</sup> pump or a K<sup>+</sup> channel in the supporting glial cells attenuates photoreceptors<sup>3</sup>. In addition to creating an optimal micro-environment, transepithelial potential differences (TEPs) are often generated to promote the functions of sensory organs. For example, the active K<sup>+</sup> transport from the perilymph to the endolymph across support cells in the mammalian auditory system<sup>4</sup> generates high driving forces that enhance the sensitivity of hair cells by increasing K<sup>+</sup> and Ca<sup>2+</sup> influx through force-gated channels. Similar designs have been found in insect mechanosensory<sup>5,6</sup> and chemosensory organs<sup>7,8</sup>, providing models to study physiological principles and components of TEP function and regulation. Many insect sensory receptor neurons are housed in a cuticular sensory organ called the sensillum. Tight junctions between support cells separate the externally facing sensillar lymph from the internal body fluid known as hemolymph<sup>9</sup>. The active concentration of K<sup>+</sup> in the dendritic sensillar lymph produces positive sensillum potentials (SP, +30~40 mV) as TEPs, which are known to sensitize sensory reception in mechanosensation<sup>10</sup> and chemosensation<sup>11,12</sup>.

Excitation of sensory neurons drains SP, accompanied by slow adaptation of the excited receptor neurons<sup>6,12</sup>. This suggests that immoderate activation of a single sensory neuron can deplete SP, which decreases the activities of neurons that utilize the potential for excitation. Each sensillum for mechanosensation and chemosensation houses multiple receptor neurons<sup>1</sup>. Therefore, overconsumption of SP by a single cell could affect the rest of the receptor neurons in the same sensillum, because the receptor neurons share the sensillum lymph. Indeed, the reduction of SP was proposed to have a negative effect on receptor neurons that are immersed in the same sensillar lymph; a dynamic lateral inhibition between olfactory receptor neurons (ORNs) occurs through “ephaptic interaction”, where SP consumption by activation of one neuron was proposed to result in hyperpolarization of an adjacent neuron, reducing its response to odorants<sup>13,14</sup>. As expected with this SP-centered model, ephaptic inhibition was reported to be mutual between *Drosophila* ORNs<sup>13,15</sup>, again because the ORNs are under the influence of a common extracellular fluid, the sensillar lymph. Such reciprocal cancellation between concomitantly excited ORNs may encode olfactory valence<sup>16</sup> rather than lead to signal attenuation of two olfactory inputs. Furthermore, depending on neuron size, the lateral inhibition between ORNs can be asymmetric, albeit yet to be bilateral; larger ORNs are more inhibitory than smaller ones<sup>13</sup>. The size dependence was suggested to be due to the differential ability of ORNs to sink SP (referred to as local field potential in the study<sup>13</sup>), probably because larger cells have more membrane surface area and cell volume to move ions to or from the sensillar lymph.

Interestingly, gustatory ephaptic inhibition was recently found to be under a genetic, but not size-aided, regulation to promote sweetness-dependent suppression of bitterness<sup>17</sup>. This is accomplished by blocking one direction of ephaptic inhibition. The hyperpolarization-activated cation current in sGRNs through the *Ih*-encoded HCN is necessary to resist the inhibition of sGRNs laterally induced by bGRN activation. Furthermore, such unilateral ephaptic inhibition is achieved against cell size gradient<sup>17</sup>. Larger bGRNs are readily suppressed by the activation of smaller sGRNs, but not vice versa. Thus, HCN is implemented to inhibit bGRNs in terms of unilateral ephaptic inhibition when a bitter chemical is concomitantly presented with strong sweetness. Here, in addition to the ephaptic interaction, we find that the same HCN expressed in sGRNs

promotes the activity of bGRNs as a means of homeostatic sensory adaptation, for which HCN prevents sGRNs from depleting SP even with the long-term exposure to the sweet-rich environment.

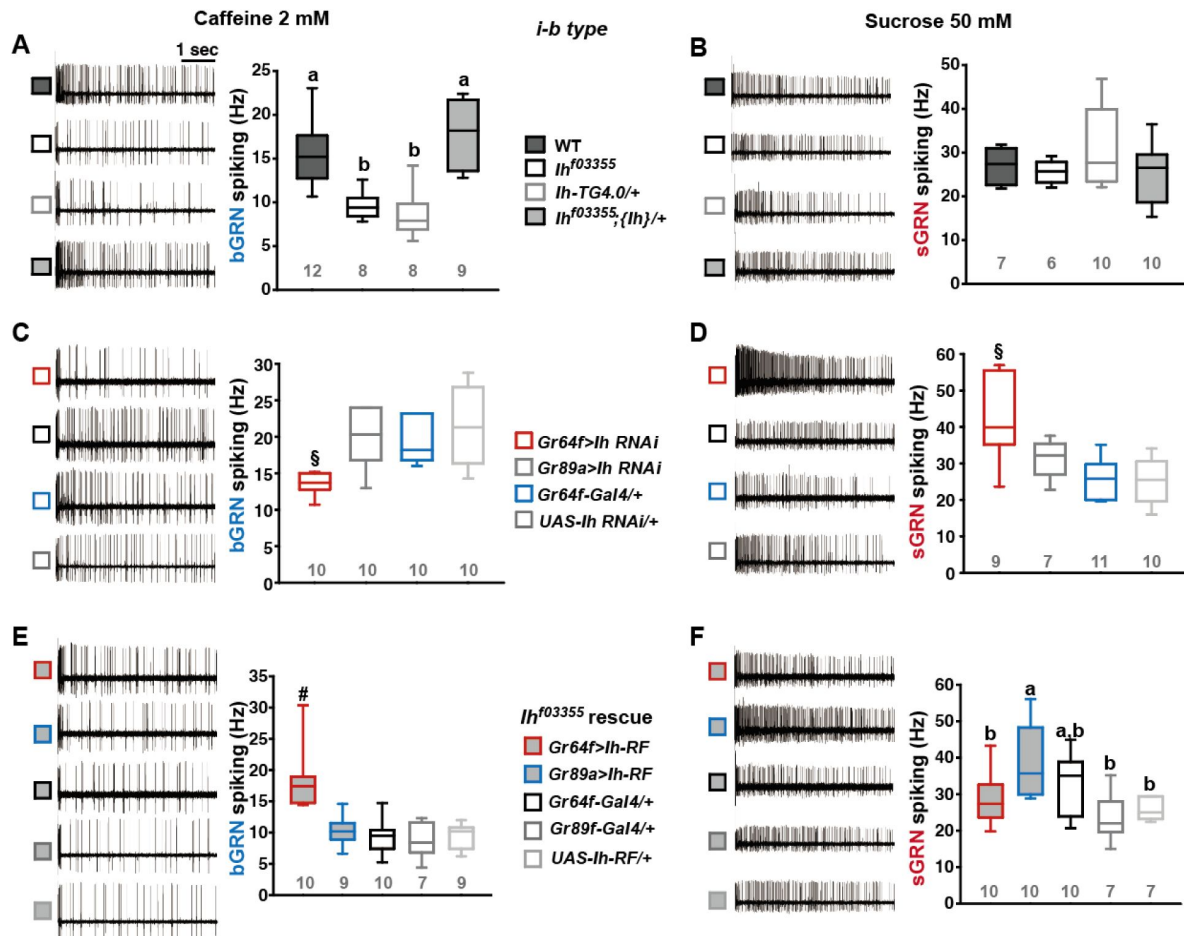
## Results

### HCN expressed in sweet-sensing GRNs is required for normal bitter GRN responses

The hair-like gustatory sensilla in the *Drosophila* labellum are categorized into L-, i-, and s-type based on their relative bristle lengths. Each sensillum contains 2 (i-type) or 4 (s- and L-type) GRNs along with a mechanosensory neuron. The i- and s-type bristle sensilla contain both an sGRN and a bGRN, while the L-type bristle sensilla contain an sGRN but no bGRN<sup>18,21</sup>. As a model of gustatory homeostasis, we mainly examined the i-type bristles using single sensillum extracellular recording<sup>22–24</sup> because of their simple neuronal composition. Compared to WT (*w<sup>1118</sup>* in a Canton S background), we observed reduced spiking responses to 2 mM caffeine in two strong loss-of-function alleles of the HCN gene, *Ih<sup>f03355</sup>*<sup>25,26</sup> and *Ih<sup>MI03196-TG4.0/+</sup>* (*Ih-TG4.0/+*)<sup>17,27</sup> (Fig. 1A). Note that *Ih-TG4.0* is homozygous lethal<sup>17</sup>. A copy of the *Ih*-containing genomic fragment {*Ih*} rescued the spiking defect in *Ih<sup>f03355</sup>*. The GRN responses to 50 mM sucrose were not altered in *Ih* mutants (Fig. 1B). Other bitter chemical compounds, berberine, lobeline, theophylline, and umbelliferone, also required *Ih* for normal bGRN responses (Fig. 1-figure supplement 1). Although we observe here that *Ih* pertains to bGRN excitability, *Ih* was previously found to be expressed in sGRNs but not bGRNs<sup>17</sup>. To test whether HCN expression in sGRNs is required for bGRN activity, GRN-specific RNAi knockdown of *Ih* was performed with either *Gr64f<sup>28</sup>* or *Gr89a-Gal4<sup>21</sup>*. *Ih* knockdown in sGRNs (*Gr64f-Gal4*), but not bGRNs (*Gr89a-Gal4*), led to reduced bGRN responses to caffeine (Fig. 1C), indicating that HCN acts in sGRNs for a normal bGRN response. Unlike the results in *Ih* mutant alleles, the spiking response of *Ih*-knockdowned sGRNs (*Gr64f* cells) to 50 mM sucrose was increased (Fig. 1D). To exclude the possibility that *Ih* is required for normal gustatory development, we temporally controlled *Ih* RNAi knockdown to occur only in adulthood, which produced similar results (Fig. 1-figure supplement 2). Such differential effects of gene disruptions and RNAi on sGRN activity will be discussed further below with additional results. Introduction of *Ih-RF* cDNA (Flybase id: FBtr0290109), which previously rescued *Ih* deficiency in other contexts<sup>17,26</sup>, to sGRNs but not bGRNs restored the decreased spiking response to 2 mM caffeine in *Ih<sup>f03355</sup>*, corroborating that sGRNs are required to express *Ih* for bGRN regulation (Fig. 1E). Interestingly, ectopic cDNA expression in bGRNs of *Ih<sup>f03355</sup>* but not in sGRNs increased the spiking response to 50-mM sucrose compared to its controls (Fig. 1F), although the same misexpression failed to raise the spiking to 2-mM caffeine. These results suggest not only that *Ih* innately expressed in sGRNs is necessary for the activity of bGRNs, but also that *Ih* expression in one GRN may promote the activity of the other adjacent GRN in *Ih*-deficient animals.

### Loss of *Ih* in sGRNs reduced the sensillum potential in the gustatory bristle sensilla

We speculated that the *Ih*-dependent lateral boosting across GRNs might involve a functional link between GRNs. Such a physiological component could be sensillum potential (SP), since the sensillar lymph is shared by all GRNs in the sensillum and SP sets the spiking sensitivity<sup>5</sup>. SP is known as a transepithelial potential between the sensillum lymph and the hemolymph, generated by active ion transport through support cells (Fig. 2A, Left). To measure SP, we repurposed the Tasteprobe pre-amplifier to record potential changes in a direct current (DC) mode (see Methods for details), which was originally devised to register action potentials from sensory neurons. With the new setting, the contact of the recording electrode with a labellar bristle induced a rise in



**Fig. 1.**

### HCN is necessary for the normal activity of bitter-sensing GRNs (bGRNs), although expressed in sGRNs.

Representative 5 sec-long traces of sensillum recording with either caffeine or sucrose at the indicated concentrations, shown along with box plots of spiking frequencies. **(A)** Caffeine-evoked bitter spiking responses of WT, the *Ih*-deficient mutants, *Ih<sup>f03355</sup>* and *Ih-TG4.0/+*, and the genomic rescue, *Ih<sup>f03355</sup>;{Ih}/+*. **(B)** Sucrose responses were similar among the genotypes tested in **(A)**. **(C)** *Ih* RNAi knockdown in sGRNs, but not bGRNs, reduced the bGRN responses to 2-mM caffeine. **(D)** *Ih* RNAi knockdown in sGRNs increased the sGRN responses to 50-mM sucrose. **(E)** Introduction of the *Ih-RF* cDNA in sGRNs, but not bGRNs, of *Ih<sup>f03355</sup>* restored the bGRN response to 2-mM caffeine. **(F)** For sucrose responses, introduction of *Ih-RF* to bGRNs increased the spiking frequency. Letters indicate statistically distinct groups (a and b): Tukey's test,  $p < 0.05$  (A), Dunn's,  $p < 0.05$  (F). §: Welch's ANOVA, Games-Howell test,  $p < 0.05$ . #: Dunn's test,  $p < 0.05$ . Numbers in gray indicate the number of tested naïve bristles, which are from at least 3 individuals.

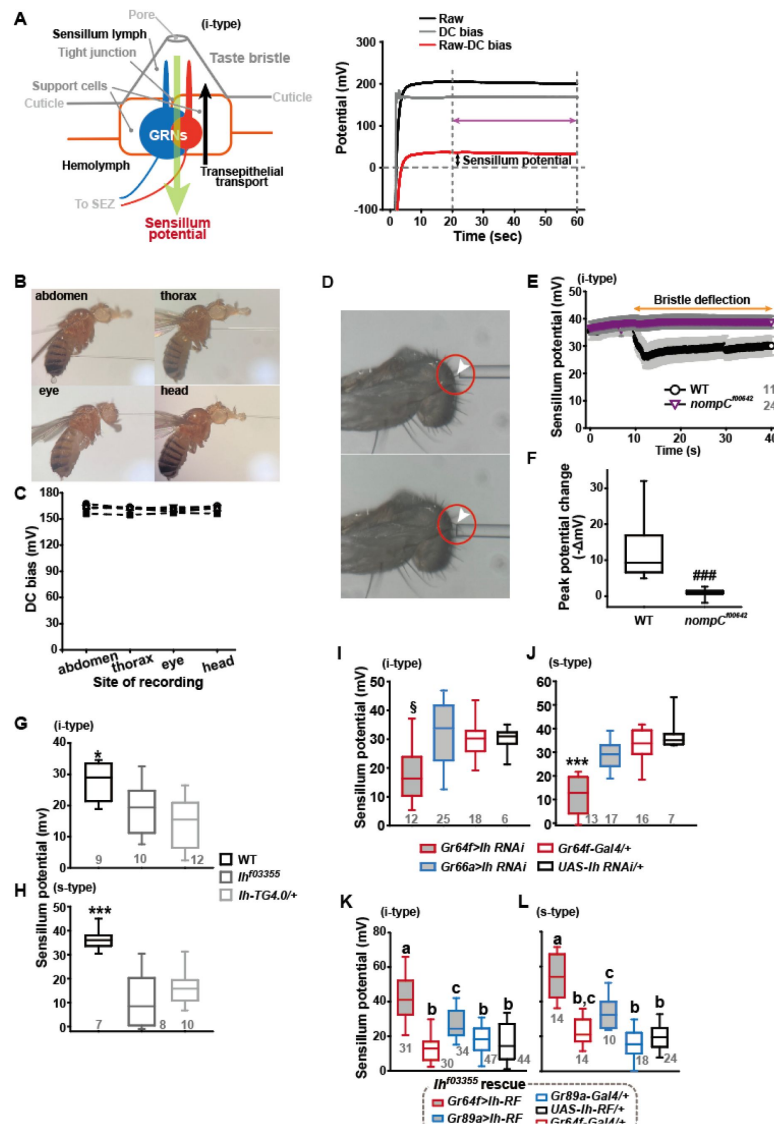
potential (Fig.2A, Right). The recording was stabilized within 20 sec, and a raw potential value was acquired as an average of the data between the time points, 20 and 60 sec after the initial contact (Fig.2A). After the examination of all the bristle sensilla of interest, the fly was impaled at the head to obtain the DC bias (also known as DC offset), which insects are known to exhibit in the body independent of SP<sup>29</sup> (Fig.2B). To examine whether the DC bias varies at different body sites, we surveyed the DC bias at four different locations of individual animals, the abdomen, thorax, eye, and head. This effort resulted in largely invariable DC bias readings (Fig.2B,C). Next, the sensillum potential was obtained by subtracting the DC bias from the raw potential value (Fig.2A). We also found that we could reduce the apparent SP by deflecting the bristle sensillum by ~45° (Fig.2D-F), activating the sensillum's mechanosensory neuron. When we performed the same experiment with *nompC*<sup>f00642</sup>, a loss-of-function allele of *nompC* that encodes a mechanosensory TRPN channel<sup>30</sup>, this reduction in SP disappeared (Fig.2D-F).

Suggesting the role of *Ih* in SP regulation, *Ih*<sup>f03355</sup> (~19 and ~10 mV for i-type and s-type sensilla, respectively) and *Ih-TG4.0/+* (~15 and ~16 mV for i-type and s-type sensilla, respectively) exhibited reduced mean SPs compared to WT in the i-type (Fig.2G) and s-type (Fig.2H) bristle sensilla (~28 mV and ~36 mV, respectively). We also examined whether the SP reduction could be attributed to the lack of *Ih* in sGRNs through GRN-specific *Ih* RNAi knockdown. This revealed that *Ih* is necessary in sGRNs for the sensilla to exhibit normal SP levels (Fig.2I,J). The SP reduction observed in both bristle types of *Ih*<sup>f03355</sup> could be fully restored by expressing the *Ih-RF* cDNA in sGRNs (*Gr64f-Gal4* cells). Mean SPs were measured to be ~42 and ~54 mV in i-type and s-type bristles, respectively (Fig.2K,L). Interestingly, ectopic expression of the cDNA in bGRNs by *Gr89a-Gal4* also significantly rescued the SP defect of *Ih*<sup>f03355</sup> to the level of mean SPs (~27 and ~33 mV in i-type and s-type bristles, respectively) comparable to those in WT. The greater extent of SP defect restoration in *Ih*<sup>f03355</sup> by *Ih-RF* expressed in sGRNs than bGRNs indicates that *Ih-RF* is more effective at upholding SP in sGRNs than in bGRNs under our experimental conditions. Furthermore, the successful rescue by *Ih-RF* in bGRNs also shows that *Ih* can regulate SP in any GRN (Fig.2K,L).

## Inactivation of sGRNs raised both bGRN activity and SP, which was reversed by *Ih* deficiency

Since it is in sGRNs that HCN regulates the bGRN responsiveness to caffeine, we suspected that the activity of sGRNs may be closely associated with the maintenance of bGRN excitability. In line with this possibility, the *Gr64af* deletion mutant, which lacks the entire *Gr64* gene locus and is severely impaired in sucrose and glucose sensing<sup>31–33</sup>, showed increased bGRN responses to various bitters in labellar gustatory bristle sensilla compared to WT (Fig.3A). Furthermore, silencing sGRNs (*Gr5a-Gal4* cells) by expressing the inwardly rectifying potassium channel, Kir2.1<sup>34</sup>, phenocopied *Gr64af* in response to 2-mM caffeine stimulating the i-type bristles (Fig.3B). This increased responsiveness of bGRNs is unlikely due to positive feedback resulting from the sGRN inactivation through the neural circuitry in the brain, because the tetanus toxin light chain (TNT) expressed in sGRNs, which blocks chemical synaptic transmission<sup>35</sup>, failed to raise bGRN activity (Fig.3C). Strikingly, when we combined the sGRN-hindering genotypes (*Gr5a>Kir2.1* and *Gr64af*) with the *Ih* alleles *Ih*<sup>f03355</sup> or *Ih-TG4.0*, we found that the sGRN inhibition-induced increase in bGRN activity in response to caffeine could be commonly relieved by the disruptions in the *Ih* gene (Fig.3B,E). This result suggests that HCN suppresses sGRN activation, while HCN expressed in sGRNs is required for unimpaired bGRN activity (Fig.1C,E). Interestingly, Kir2.1-induced inactivation of sGRNs (*Gr64f-Gal4* cells) dramatically increased the mean SP of the i-type bristles to ~53 mV, compared to ~29 and ~35 mV of *Gal4* and *UAS* controls, respectively (Fig.3D), and the impairment of sucrose-sensing in the *Gr64af* mutants also resulted in increases of mean SPs (Fig.3F, ~56 and ~53 mV in the i- and s-bristles of *Gr64af*, compared to ~30 and ~36 mV of WT, respectively). Thus, inactivating sGRNs in two different ways increased SP in the i- and s-type gustatory bristles, similar to the effect on bGRN activity described earlier. Such repeated parallel shifts of bGRN activity and SP were again obtained in the combined genotypes between *Gr64af*





**Fig. 2.**

### Sensillum potential (SP) is reduced in HCN-deficient animals.

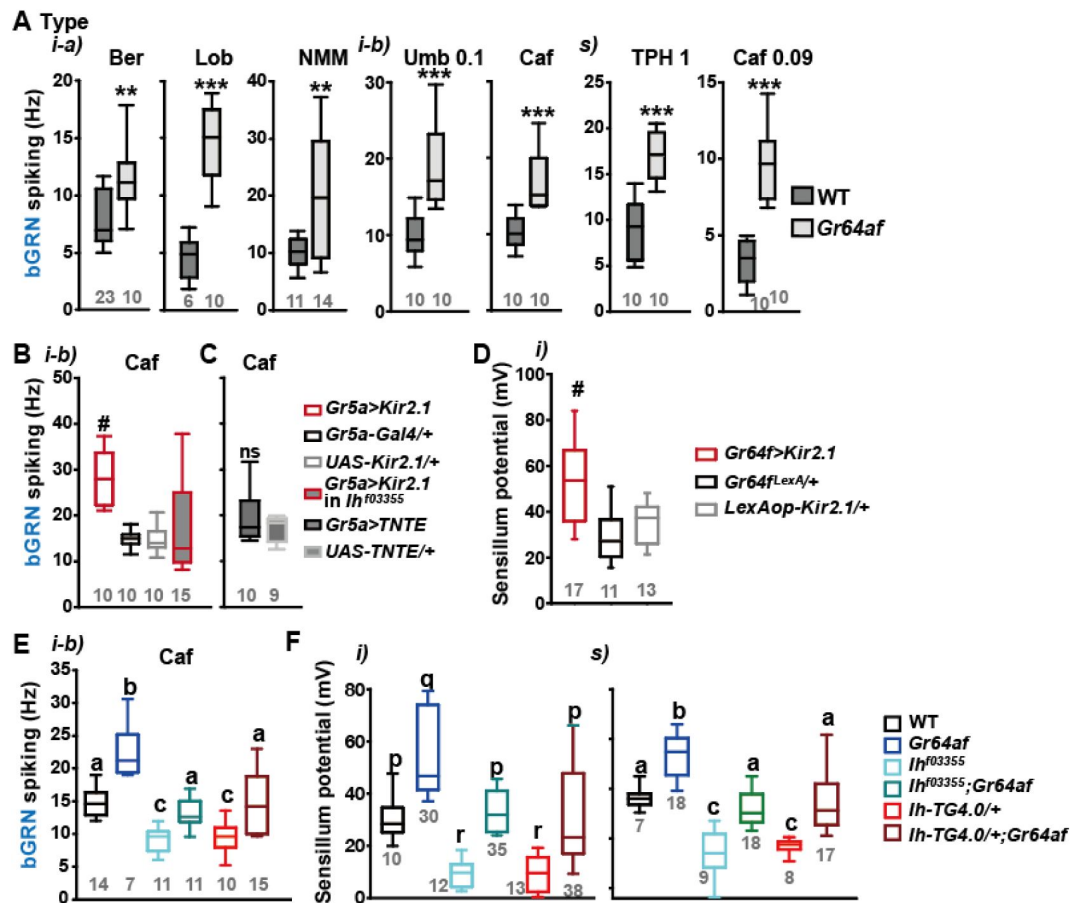
(A) Schematic diagram illustrating the sensillum potential in the taste bristle sensilla (Left). Black upward arrow indicates ion transport by pumps and transporters in support cells from the hemolymph to the sensillum lymph, which are physiologically separated by tight junctions between support cells. The resulting transcellular disparity of ions leads to a positive sensillum potential. Representative traces of potentials measured to evaluate SP (Right). Raw: the potential reading upon the contact of the recording electrode with the sensillum bristle tip (black). DC bias: the potential reading upon impalement of the head by the recording electrode (gray). Red line indicates the difference between raw and DC bias, which represents the sensillum potential. The values resulting from the subtraction of the data between 20 to 60 sec after the initial contact (time indicated by the purple double headed arrow) were averaged to determine SP. (B) Photographs of impaled flies for DC bias determination at indicated sites. (C) DC bias values obtained from indicated body parts. There is no statistical significance between the body sites (ANOVA Repeated Measures). (D) Photos before (top) and after (bottom) deflection of an i-type bristle. (E) Sensillum potential traces as a function of time from WT and *nompC*<sup>00642</sup>. Bristle bending started at 10 sec, and the duration is marked by an orange double headed arrow. (F) The peak SP changes of WT and *nompC*<sup>00642</sup> were compared. (G and H) SP was reduced in i- (G) and s-type (H) bristles of the indicated *Ih*-deficient mutants, relative to WT. (I and J) *Ih* RNAi in sGRNs reduced SPs of the i- and s-type bristles. (K and L) The SP of *Ih*<sup>03355</sup> was restored by *Ih*-RF expression in GRNs (red for sGRNs, blue for bGRNs). ###: Dunn's, *p* < 0.001. \* and \*\*\*: Tukey's, *p* < 0.05 and *p* < 0.001, respectively. S: Welch's ANOVA, Games-Howell test, *p* < 0.05. Letters indicate statistically distinct groups: Tukey's test, *p* < 0.05. Numbers in gray indicate the number of naive bristles tested in at least three animals.

gene locus and is severely impaired in sucrose and glucose sensing<sup>31,33</sup>, showed increased bGRN responses to various bitters in labellar gustatory bristle sensilla compared to WT (**Fig.3A**). Furthermore, silencing sGRNs (*Gr5a-Gal4* cells) by expressing the inwardly rectifying potassium channel, Kir2.1<sup>34</sup>, phenocopied *Gr64af* in response to 2-mM caffeine stimulating the i-type bristles (**Fig.3B**). This increased responsiveness of bGRNs is unlikely due to positive feedback resulting from the sGRN inactivation through the neural circuitry in the brain, because the tetanus toxin light chain (TNT) expressed in sGRNs, which blocks chemical synaptic transmission<sup>35</sup>, failed to raise bGRN activity (**Fig.3C**). Strikingly, when we combined the sGRN-hindering genotypes (*Gr5a>Kir2.1* and *Gr64af*) with the *Ih* alleles *Ih<sup>f03355</sup>* or *Ih-TG4.0*, we found that the sGRN inhibition-induced increase in bGRN activity in response to caffeine could be commonly relieved by the disruptions in the *Ih* gene (**Fig.3B,E**). This result suggests that HCN suppresses sGRN activation, while HCN expressed in sGRNs is required for unimpaired bGRN activity (**Fig.1C,E**). Interestingly, Kir2.1-induced inactivation of sGRNs (*Gr64f-Gal4* cells) dramatically increased the mean SP of the i-type bristles to ~53 mV, compared to ~29 and ~35 mV of *Gal4* and *UAS* controls, respectively (**Fig.3D**), and the impairment of sucrose-sensing in the *Gr64af* mutants also resulted in increases of mean SPs (**Fig.3F**, ~56 and ~53 mV in the i- and s-bristles of *Gr64af*, compared to ~30 and ~36 mV of WT, respectively). Thus, inactivating sGRNs in two different ways increased SP in the i- and s-type gustatory bristles, similar to the effect on bGRN activity described earlier. Such repeated parallel shifts of bGRN activity and SP were again obtained in the combined genotypes between *Gr64af* and *Ih<sup>f03355</sup>* or *Ih-TG4.0/+* (**Fig.3F**); the SP increased in *Gr64af* descended to WT levels when combined with *Ih<sup>f03355</sup>* and *Ih-TG4.0/+*, similar to what occurred with bGRN activity in *Gr64af* (**Fig.3E**). These results suggest that *Ih* gene expression suppresses sGRNs, upholding both bGRN activity and SP, similar to the genetic alterations that reduce sGRN activity.

Water GRNs are co-housed with sGRNs in L-type bristles in the labellum, responding to hypo-osmolarity with the aid of *ppk28* and promoting water drinking<sup>36,37</sup>. We tested whether *Ih*-dependent SP regulation occurs in these bristles to maintain the sensitivity of water GRNs by using a low concentration of the electrolyte tricholine citrate (TCC) at 0.1 mM. Interestingly, L-type bristles of *Ih<sup>f03355</sup>* showed reduced spike frequencies in response to this hypo-osmolar electrolyte solution compared to WT (Fig. 3-figure supplement 1A). This reduction was restored in the genetic rescue line. Additionally, SP in these bristles was increased in *Gr64af* but decreased in the two *Ih* alleles, and the combination of the *Gr64* and *Ih* mutations restored SP to the level of WT (Fig. 3-figure supplement 1B), as observed with other sensillar bristles above. Finally, *Ih-RF* restored SP in *Ih<sup>f03355</sup>* when expressed in sGRNs but not bGRNs, as expected from the absence of bGRNs in L types (Fig. 3-figure supplement 1C). Thus, *Ih*-dependent SP regulation is universal in all bristle sensilla of the labellum and likely important for the function of GRNs neighboring sGRNs.

## HCN delimits excitability of HCN-expressing GRNs, and increases SP

By misexpressing *Ih-RF* in bGRNs of WT flies, we investigated how HCN physiologically controls HCN-expressing GRNs (**Fig.4A**). The genetic controls, *Gr89a-Gal4/+* and *UAS-Ih-RF/+*, exhibited mutually similar dose dependencies saturated at 2- and 10-mM caffeine, revealing the maximal caffeine responses at these concentrations. Interestingly, the ectopic expression reduced bGRN activity at these high caffeine concentrations (**Fig.4A**). The flattened dose dependence suggests that ectopically expressed HCN suppresses strong excitation of bGRNs. In contrast, sGRNs were upregulated by the misexpression of *Ih* in bGRNs with increased spiking in response to 10- and 50-mM sucrose (**Fig.4B**), implying that *Ih* increases the activity of the neighboring GRN by reducing that of *Ih*-expressing GRNs. On the other hand, the *Ih-RF*-overexpressing sGRNs in *Gr64f-Gal4* cells significantly decreased only the response 5 sec after contacting 50-mM sucrose (**Fig.4C**, the second 5-sec bin, **Fig. 4**-figure supplement 1), probably because of native HCN preoccupying WT sGRNs. Although bGRNs were repressed by misexpressing *Ih-RF*, the mean SPs increased to ~40 and ~37 mV in the i- and s-type bristles, respectively, compared to controls with mean SPs of 22-25



**Fig. 3.**

### Inactivation of sGRNs raises bGRN activity and SP, both of which are reversed by *Ih* deficiency.

(A) The bGRN spiking was increased in response to the indicated bitter tastants in *Gr64af* mutants impaired in sucrose and glucose sensing. Ber: 0.5, Lob: 0.5, NMM: 2, Caf: 2 (i-type) and 0.09 (s-type), Umb: 0.1, TPH: 1 mM. \* and \*\*\*: Student's t-test,  $p < 0.01$  and  $p < 0.001$ , respectively. (B and C) Silencing by Kir2.1 (B), but not blocking chemical synaptic transmission (C), in sGRNs increased the spiking of bGRNs stimulated by 2-mM caffeine, which was reversed in *Ih<sup>f03355</sup>* (B). #: Dunn's,  $p < 0.05$ . (D) Silencing sGRNs by Kir2.1 increased SP. #: Dunn's,  $p < 0.05$ . (E) The increased bGRN spiking in *Gr64af* was restored to WT levels by *Ih* deficiencies. Letters indicate significantly different groups (Tukey's,  $p < 0.05$ ). Caffeine 2 mM was used (B, C and E). (F) Regardless of bristle type, SP was increased upon sGRN inactivation, which was reduced by *Ih* deficiencies. p-r: Dunn's test,  $p < 0.05$ . a-c: Welch's ANOVA, Games-Howell test,  $p < 0.05$ . Numbers in gray indicate the number of naïve bristles in at least 3 animals.

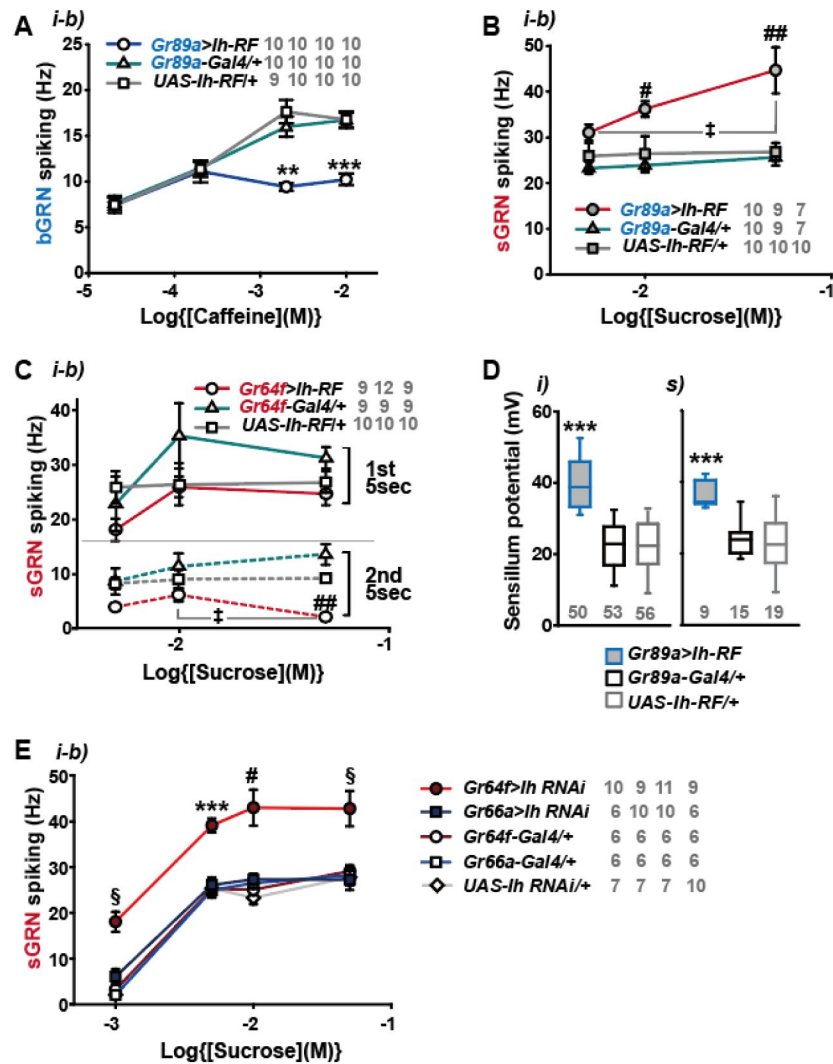


mV (Fig.4D). These results from misexpression experiments corroborate the postulation that sGRNs are suppressed by expressing HCN. To confirm that sGRNs are suppressed by native HCN, the impact of GRN-specific *Ih* RNAi knockdown on sGRNs was quantitatively evaluated (Fig.4E). *Ih* RNAi in sGRNs (*Gr64f-Gal4* cells) led to increased mean spiking frequencies by ~10 Hz in response to 1-, 5-, and 10- mM as well as 50-mM sucrose compared to *Ih* RNAi in bGRNs (*Gr66a-Gal4* cells) and genetic controls, highlighting the extent to which HCN natively expressed in sGRNs suppresses sGRN excitability. In contrast, SP, necessary for GRN sensitization, was observed above to be reduced by *Ih* RNAi in sGRNs but not bGRNs (Fig.2I,J). Thus, these data suggest that HCN innately reduces the spiking frequencies of sGRNs even at relatively low sucrose concentrations, 1, and 5 mM. This is similar to the suppressive effect of *Ih-RF* misexpressed in bGRNs at relatively high caffeine concentrations, but differs in that the misexpression did not alter bGRN activity in response to low caffeine concentrations, 0.02 and 0.2 mM (Fig.4A), implying a complex cell-specific regulation of GRN excitability.

## Sweetness in the food leads to reduction of SP, bGRN activity, and bitter avoidance in *Ih*-deficient animals

Typically, we performed extracellular recordings on flies 4-5 days after eclosion, during which they were kept in a vial with fresh regular cornmeal food containing ~400 mM D-glucose. The presence of sweetness in the food would impose strong and frequent stimulation of sGRNs for extended period, potentially requiring the delimitation of sGRN excitability for the homeostatic maintenance of gustatory functions. To investigate this possibility, we fed WT and *Ih*<sup>f03355</sup> flies overnight with either non-sweet sorbitol alone (200 mM) or a sweet mixture of sorbitol (200 mM) + sucrose (100 mM). Although sorbitol is not sweet, it is a digestible sugar that provides *Drosophila* with calories<sup>38</sup>. We found that the sweet sucrose medium significantly reduced caffeine-induced bGRN responses in both genotypes compared to the sorbitol only medium, but *Ih*<sup>f03355</sup> bGRN spike frequencies were decreased to the level significantly lower than WT (Fig.5A), as seen above with the cornmeal food (Figs.1A,C and 3E). This suggests that the reduced bGRN activity in the mutants may result from prolonged sGRN excitation. The SP reduction was induced by 1-hr incubation with the sweet sucrose medium in both WT and *Ih*<sup>f03355</sup> to a similar degree. However, the *Ih* mutant showed a more severe depletion of SPs compared to WT after 4 hrs of sweet exposure (Fig.5B) as observed with the cornmeal food (Figs.2 and 3F). Even on the sorbitol food, the SP in *Ih*<sup>f03355</sup> was significantly decreased compared to WT. This may be attributed to the loss of HCN, which is known to stabilize the resting membrane potential<sup>39</sup>. Following overnight sweet exposure, SPs of WT and *Ih*<sup>f03355</sup> were recovered to similar levels after 1-hr incubation with sorbitol only food. However, it was after 4 hrs on the sorbitol food that the two lines exhibited SP levels similar to those achieved by overnight incubation with sorbitol only food (Fig. 5B). These results indicate that SP depletion by sweetness is a slow process, and that the dysregulated reduction and recovery of SPs in *Ih*<sup>f03355</sup> manifest only after long-term conditioning with and without sweetness, respectively.

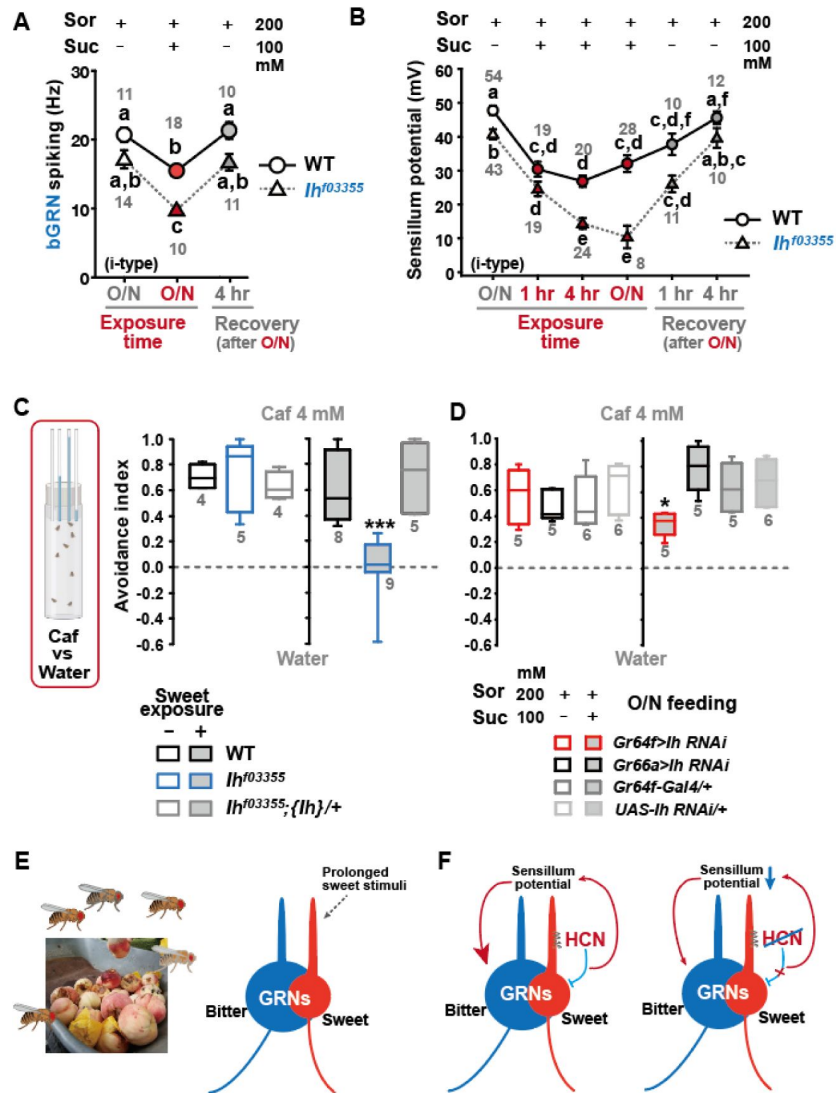
To assess the behavioral implications of HCN-assisted preservation of SP and bGRN activity, flies were exposed long-term to sweetness on a regular sweet cornmeal diet (sweet exposure-positive), and then subjected to a CAFE with an 8-hr choice between water and 4 mM caffeine solution. Note that sucrose was not used in CAFE, because the presence of sweet stimuli was shown to suppress bGRNs<sup>17</sup>. Indicative of reduced bitter sensitivity, *Ih*<sup>f03355</sup> flies showed dramatically decreased caffeine avoidance, relative to WT (Fig.5C). In contrast, when flies were removed from the cornmeal food for 20 hrs, both WT and *Ih*<sup>f03355</sup> showed similarly robust bitter avoidance. The defect observed in the *Ih* mutant on the sweet cornmeal diet could be rescued by reintroducing a genomic fragment covering the *Ih* locus (*Ih*). These results were recapitulated with other bitters, lobeline and theophylline (Fig. 5-figure supplement 1). To examine whether caffeine avoidance requires *Ih* expression in sGRNs, CAFE was performed with GRN-specific RNAi knockdown of *Ih*. For the RNAi experiments, flies were kept overnight on either the non-sweet diet with sorbitol (200



**Fig. 4.**

### HCN suppresses HCN-expressing GRNs and increases SP.

(A) HCN misexpressed in bGRNs flattened the dose dependence to caffeine. (B) HCN ectopically expressed in bGRNs elevates sGRN responses to sucrose. (C) Overexpression of HCN in sGRNs reduced the sGRN responses to sucrose 5 sec after the initial contact. (D) *Ih* misexpression in bGRNs increased SP in i- and s-type bristles, which correlates with laterally increased sGRN activity (B). (E) *Ih* RNAi knockdown in sGRNs (*Gr64f-Gal4* cells) dramatically elevates spiking frequencies in response to 1-, 5-, 10-, and 50-mM sucrose. \*, \*\*, and \*\*\*: Tukey's,  $p < 0.05$ ,  $p < 0.01$ , and  $p < 0.001$ , respectively (A,D,E). # and ##: Dunn's,  $p < 0.05$  and  $p < 0.01$  between genotypes, respectively (B,C,E). †: Dunn's,  $p < 0.05$  between responses to different sucrose concentrations (B,C). §: Welch's ANOVA, Games-Howell test,  $p < 0.05$  (E). The numbers in grey indicate the number of tested naïve bristles in at least three animals.



**Fig. 5.**

### Sweetness in the diet decreases SP, bGRN activity, and bitter avoidance.

(A) Sweetness in the media reduced the 2-mM caffeine-evoked bGRN spiking, which was fully recovered in four-hour incubation with sorbitol only food.  $Ih^{f03355}$  was affected by the type of the media more severely than WT. O/N: overnight incubation with sorbitol only (grey) or sucrose food (red). (B) The SP of  $Ih^{f03355}$  bristle sensilla showed dysregulated reduction after four-hour and overnight incubation on sweet media. These reductions started to be recovered in one-hour feeding and were nearly fully recovered in four-hour feeding on the indicated sorbitol only food. (C) Caffeine (Caf) avoidance was assessed with capillary feeder assay (CAFE).  $Ih$  is required for robust caffeine avoidance for flies maintained on sweet cornmeal food (sweet exposure +: filled boxes).  $Ih^{f03355}$  flies avoided 4 mM caffeine like WT flies when separated from sweet food for 20 hrs (blank boxes). (D)  $Ih$  RNAi knockdown in sGRNs ( $Gr64f-Gal4$ ) but not bGRNs ( $Gr66a-Gal4$ ) led to relatively poor avoidance to caffeine after feeding on the sweet diet with sucrose. Suc: sucrose, and Sor: sorbitol Letters indicate statistically distinct groups: a-f, Dunn's,  $p < 0.05$  (A,B). \* and \*\*\*: Tukey's,  $p < 0.05$  and  $< 0.001$ , respectively. (E) Illustration depicting the flies' sweet feeding niche in overripe fruit (Left), leading to prolonged exposure of sGRNs to the sweetness (Right). (F) A schematic model of gustatory homeostasis in *Drosophila* bristle sensilla. Despite the prolonged sweetness in the environment robustly and frequently stimulating sGRNs, the sGRN activity is moderated by HCN to preserve the sensillum potential, which is required for normal bGRN responsiveness (Left). When HCN in sGRNs is incapacitated, sGRNs can become overly excited by sweetness of overripe fruit and deplete the sensillum potential, resulting in decreased bGRN activity and bitter avoidance (Right).

mM) or the sweet diet with additional sucrose (100 mM). *Ih* knockdown in sGRNs, but not bGRNs, led to a deficit in the avoidance only when the flies were on the sweet diet, indicating that HCN expression in sGRNs is necessary for robust caffeine avoidance in a sweet environment (Fig. 5D). Therefore, the sweetness of the diet can compromise the function of bGRNs co-housed with sGRNs in the same sensilla, which is mitigated by HCN expression in sGRNs. Such a role of HCN is essential for bitter avoidance of flies, considering their likely prolonged exposure to sweetness in their natural habitat of overripe fruit (Fig. 5E).

## Discussion

Our results provide multiple lines of evidence that HCN suppresses HCN-expressing GRNs, thereby sustaining the activity of neighboring GRNs within the same sensilla (Fig. 5F). We propose that this modulation occurs by restricting SP consumption through HCN-dependent neuronal suppression rather than via chemical and electrical synaptic transmission. The lack of increased bGRN activity with TNT expression in sGRNs, coupled with the increase observed with Kir2.1 expression (Fig. 3B,C), indicates minimal involvement of synaptic vesicle-dependent transmission. The possibility of a neuropeptide-dependent mechanism is unlikely, given our ectopic gain-of-function studies (Fig. 4). To explain the misexpression results with neuropeptide pathways, both s- and bGRNs must be equipped with the same set of a neuropeptide/receptor system, which is incompatible with the inverse relationship between the two GRNs in excitability observed in Fig. 1C,D, Fig. 1-figure supplement 2B, and Fig. 3. Furthermore, this inverse relationship argues against electrical synapses through gap junctions, which typically synchronize the excitability of pre- and postsynaptic neurons. Therefore, our findings propose an unconventional mechanism of neuronal interaction.

HCNs are encoded by four different genes in mammals<sup>39,40</sup>, and are known to be present in mammalian sensory receptor cells. In cochlear hair cells, HCN1 and HCN2 were reported to form a complex with a stereociliary tip-link protein<sup>41</sup>, while in vestibular hair cells, HCN1 is essential for normal balance<sup>42</sup>. HCN1 was also immunostained in cone and rod photoreceptors, as well as retinal bipolar, amacrine, and ganglion neurons, with deletion of the encoding gene resulting in prolonged light responses<sup>43</sup>. A subset of mouse taste cells was labeled for HCN1 and HCN4 transcripts and proteins<sup>44</sup>, similar to our observation of selective HCN expression in *Drosophila* GRNs. HCN2 is expressed in small nociceptive neurons that mediate diabetic pain<sup>45</sup>. However, the precise roles of HCNs in regulating these respective sensory physiologies remain to be elucidated.

HCN is well-known for its ‘funny’ electrophysiological characteristics, stabilizing the membrane potential<sup>39,40</sup>. As a population of HCN channels remains open at the resting membrane potential, HCN serves to suppress neuronal excitation in two ways. First, it increases the inward current required to depolarize the membrane and trigger action potentials, owing to the low membrane input resistance resulting from the HCN-dependent passive conductance. Second, the closing of HCN induced by membrane depolarization counteracts the depolarization, since the reduction of the standing cation influx through HCN is hyperpolarizing. Conversely, HCNs also allow neurons to resist membrane hyperpolarization because the hyperpolarization activates HCNs to conduct depolarizing inward currents. Consequently, HCN channels effectively dampen fluctuations in membrane potential, whether they lead to depolarization or hyperpolarization. Our findings in this study align with the former property of HCNs, as *Drosophila* HCN is essential for moderating sGRN excitation to preserve SP and bGRN activity when flies inhabit in sweet environments. On the other hand, our previous study showed that HCN-dependent resilience to hyperpolarizing inhibition of sGRNs lateralizes gustatory ephaptic inhibition to dynamically repress bGRNs, when exposed to strong sweetness together with bitterness<sup>17</sup>. Thus, depending on the given feeding contexts, the electrophysiological properties of HCN in sGRNs lead to playing dual roles with opposing effects in regulating bGRNs. The stabilization of membrane potential by

HCNs was reported to decrease the spontaneous activity of neurons, as evidenced by miniature postsynaptic currents suppressed by presynaptic HCNs<sup>46</sup>. In this regard, the lower SP observed in the *Ih*<sup>03355</sup> labellar bristles than that of WT, even on the nonsweet sorbitol food (**Fig.5B**), may be attributed to the more facile fluctuations in resting membrane potential which could regulate the consumption of SP (further discussion below).

Cell-specific knockdown of *Ih* in sGRNs led to increased sGRN responses to 50-mM sucrose (**Fig.1D**), although disruptions of the *Ih* locus did not (**Fig.1B**). This inconsistency may stem from differences between alleles and the RNAi knockdown in residual *Ih* expression or in *Ih*-deficient sites. The lack of *Ih* in sGRNs can induce two different effects in neuronal excitation: 1) easier depolarization of sGRNs due to the loss of standing HCN currents at rest (as suggested in **Fig.4E**) and 2) a decrease of receptor-mediated inward currents, expected due to SP reductions (**Fig.2G-L**). Assuming that some level of HCN expression may persist in RNAi knockdowns compared to mutants, these opposing effects on sGRN excitability may largely offset each other in response to 50-mM sucrose in the *Ih* mutants, but not in the knockdowned flies. The ectopic introduction of *Ih-RF* into bGRNs of *Ih*<sup>03355</sup> significantly increased the mean SP compared to the control genotypes (**Fig.2K,L**), leaving sGRNs devoid of functional *Ih*. This genotype allows the examination of sGRNs lacking *Ih*, with SP unimpaired, which is supposed to reflect the net effect of *Ih* on sGRN excitability excluding the influence from reduced SP. Interestingly, the ectopic rescue resulted in elevated firing responses to 50-mM sucrose compared to the cDNA rescue in sGRNs (**Fig.1F**), a proper control with *Ih* expression and SP both unimpaired. On the other hand, the differing sites of *Ih* deficiency might create the inconsistency. The protein trap reporter *Ih-TG4.0-Gal4* previously showed widespread expression of HCN in the labellum, including non-neuronal cells, implying the possibility of unknown bGRN-regulating HCN-dependent mechanisms, potentially harbored in nonneuronal cells. Overall, our cell-specific loss-of-function and gain-of-function studies advocate that HCN suppresses HCN-expressing GRNs, which thereby increases SP to promote the activity of the neighboring GRNs.

Only the dendrites of GRNs face the sensillar lymph, separated from the hemolymph by tight junctions between support cells<sup>9</sup>. The inward current through the ion channels that respond to sensory reception in the dendrites is thought to be a major sink for SP<sup>5,12</sup>, consistent with the incremented SP in the *Gr64af* mutant lacking the sucrose-sensing molecular receptor<sup>17,31</sup>. Based on these points, it was somewhat unexpected that the membrane potential regulator HCN preserved SP, yet implying that the sensory signaling in the dendrite is likely under voltage-dependent control. In line with HCN, shifting the membrane potential toward the K<sup>+</sup> equilibrium by overexpressing Kir2.1 in sGRNs upregulated bGRN activity and SP (**Fig.3**), corroborating that the membrane potential in sGRNs is a regulator of the sensory signaling cascade in the dendrites. Note that the sensillum lymph contains high [K<sup>+</sup>]<sup>5,7</sup>, which would not allow strong inactivation of sGRNs and SP increases if Kir2.1 operates mostly in the dendrites. The increases in SP, coinciding with the apparent silencing of sGRNs by Kir2.1<sup>17</sup>, propose that lowering the membrane potential in the soma and the axon suppresses the consumption of SP probably by inhibiting the gustatory signaling-associated inward currents in the dendrite. Para, the *Drosophila* voltage-gated sodium channel, was reported to be localized in the dendrites of mechanosensitive receptor neurons in *Drosophila* chordotonal organs<sup>47</sup>. Similarly, *Drosophila* voltage-gated calcium channels have been studied in dendrites<sup>48-50</sup>, implying that membrane potential may be an important contributor to the sensory signaling in dendrites.

There are ~14,500 hair cells in the human cochlea at birth<sup>51</sup>. These hair cells share the endolymph in the scala media (cochlear duct), representing a case of TEP shared by a large group of sensory receptor cells. Since HCNs were found to be unnecessary for mechanotransduction itself in the inner ear<sup>52</sup>, they may play a regulatory role in fine-tuning the balance between the endocochlear potential maintenance and mechanotransduction sensitivity for hearing, as in the *Drosophila* gustatory system. Multiple mechanosensory neurons are found to be co-housed also in *Drosophila* mechanosensory organs such as hair plates and chordotonal organs<sup>5</sup>. Given that



each mechanosensory neuron is specifically tuned to detect different mechanical stimuli such as the angle, velocity, and acceleration of joint movement<sup>53</sup>, some elements of these movements may occur more frequently and persistently than others in a specific ecological niche. Such biased stimulation would require HCN-dependent moderation to preserve the sensitivity of other mechanoreceptors sharing the sensillar lymph. We showed that ectopic expression of *Ih* in bGRNs also upheld SP and the activity of the neighboring sGRNs, underscoring the independent capability of HCN in SP preservation. Despite such an option available, the preference for sGRNs over bGRNs in HCN-mediated taste homeostasis implies that *Drosophila melanogaster* may have ecologically adapted to the high sweetness<sup>54</sup> prevalent in their feeding niche, such as overripe or fermented fruits<sup>55</sup>. It would be interesting to investigate whether and how respective niches of various insect species differentiate the HCN expression pattern in sensory receptor neurons for ecological adaptation.

In this report, we introduce a peripheral coding design for feeding decisions that relies on HCN. HCN operating in sGRNs allows uninterrupted bitter avoidance, even when flies reside in sweet environments. This is achieved in parallel with an ephaptic mechanism of taste interaction by the same HCN in sGRNs, whereby bitter aversion can be dynamically attenuated in the simultaneous presence of sweetness<sup>17</sup>. Further studies are warranted to uncover similar principles of HCN-dependent adaptation in other sensory contexts. It would also be interesting to explore whether the role of HCN in the sensory adaptation consistently correlates with lateralized ephaptic inhibition between sensory receptors, given that sensory cells expressing HCN can resist both depolarization and hyperpolarization of the membrane.

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Investigation: MHL, KJK, KMJ, SHP, KL

Visualization: MHL, KJK, SHP, KL

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Writing – review & editing: JYK, KMJ, SHP

## Competing interests

Authors declare that they have no competing interests.

## Data and materials availability

All data are available in the main text or the supplementary materials.

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## Editors

Reviewing Editor

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## Reviewer #1 (Public Review):

Summary:

This study identifies new types of interactions between *Drosophila* gustatory receptor neurons (GRNs) and shows that these interactions influence sensory responses and behavior. The authors find that HCN, a hyperpolarization-activated cation channel, suppresses the activity of GRNs in which it is expressed, preventing those GRNs from depleting the sensillum potential, and thereby promotes the activity of neighboring GRNs in the same sensilla. HCN is expressed in sugar GRNs, so HCN dampens excitation of sugar GRNs and promotes excitation of bitter GRNs. Impairing HCN expression in sugar GRNs depletes the sensillum potential and decreases bitter responses, especially when flies are fed on a sugar-rich diet, and this leads to decreased bitter aversion in a feeding assay. The authors' conclusions are supported by genetic manipulations, electrophysiological recordings, and behavioral assays.

Strengths:

(1) Non-synaptic interactions between neurons that share an extracellular environment (sometimes called "ephaptic" interactions) have not been well-studied, and certainly not in the insect taste system. A major strength of this study is the new insight it provides into how these interactions can impact sensory coding and behavior.

(2) The authors use many different types of genetic manipulations to dissect the role of HCN in GRN function, including mutants, RNAi, overexpression, ectopic expression, and neuronal silencing. Their results convincingly show that HCN impacts the sensillum potential and has both cell-autonomous and nonautonomous effects that go in opposite directions. Temporally controlled RNAi experiments suggest that the effect is not due to developmental changes. There are a couple of conflicting or counterintuitive results, but the authors discuss potential explanations.

(3) Experiments comparing flies raised on different food sources suggest an explanation for why the system may have evolved the way that it did: when flies live in a sugar-rich environment, their bitter sensitivity decreases, and HCN expression in sugar GRNs helps to counteract this decrease. New experiments in the revised paper show the timecourse of how sugar diet affects GRN responses and sensillum potential.

#### Weaknesses/Limitations:

(1) The RNAi Gal80ts experiment only compares responses of experimental flies housed at different temperatures without showing control flies (e.g. Gal4/+ and UAS/+ controls) to confirm that observed differences are not due to nonspecific effects of temperature. Certainly temperature cannot account for sugar and bitter GRN firing rates changing in opposite directions, but it may have some kind of effect.

(2) The experiments where flies are put on sugar vs. sorbitol food show that the diet clearly affects GRN responses and sensillum potential, even for food exposures as short as 1-4 hours, but it is not clear to what extent the GRNs in the labellum are being stimulated during those incubation periods. The flies are most likely not feeding over a 1 hour period if they were not starved beforehand, in which case it is not clear how many times the labellar GRNs would contact the food substrate.

(3) The authors mention that HCN may impact the resting potential in addition to changing the excitability of the cell through various mechanisms. It would be informative to record the resting potential and other neuronal properties, but this is very difficult for GRNs, so the current study is not able to determine exactly how HCN affects GRN activity.

<https://doi.org/10.7554/eLife.96602.2.sa3>

#### Reviewer #2 (Public Review):

##### Summary:

In this manuscript, the authors show that HCN loss-of-function mutation causes a decrease in spiking in bitter GRNs (bGRN) while leaving sweet GRN (sGRN) response in the same sensillum intact. They show that a perturbation of HCN channels in sweet-sensing neurons causes a similar decrease while increasing the response of sugar neurons. They were also able to rescue the response by exogenous expression. Ectopic expression of HCN in bitter neurons had no effect. Next, they measure the sensillum potential and find that sensillum potential is also affected by HCN channel perturbation. These findings lead them to speculate that HCN in sGRN increases sGRN spiking, which in turn affects bGRNs. To test this idea, they carried out multiple perturbations aimed at decreasing sGRN activity. They found that reducing sGRN activity by either using receptor mutant or by expressing Kir (a K<sup>+</sup> channel) in sGRN increased bGRN responses. These responses also increase the sensillum potential. Finally, they show that these changes are behaviorally relevant as conditions that increase sGRN activity decrease avoidance of bitter substances.

##### Strengths:

There is solid evidence that perturbation of sweet GRNs affects bitter GRN in the same sensillum. The measurement of transsynaptic potential and how it changes is also interesting and supports the author's conclusion

Weaknesses:

The ionic basis of how perturbation in GRN affects the transepithelial potential, which in turn affects the second neuron, is unclear.

<https://doi.org/10.7554/eLife.96602.2.sa2>

Reviewer #3 (Public Review):

Ephaptic inhibition between neurons housed in the same sensilla has been long discovered in flies, but the molecular basis underlying this inhibition is underexplored. Specifically, it remains poorly understood which receptors or channels are important for maintaining the transepithelial potential between the sensillum lymph and the hemolymph (known as the sensillum potential), and how this affects the excitability of neurons housed in the same sensilla.

Lee et al. used single-sensillum recordings (SSR) of the labellar taste sensilla to demonstrate that the HCN channel, *Ih*, is critical for maintaining sensillum potential in flies. *Ih* is expressed in sugar-sensing GRNs (sGRNs) but affects the excitability of both the sGRNs and the bitter-sensing GRNs (bGRNs) in the same sensilla. *Ih* mutant flies have decreased sensillum potential, and bGRNs of *Ih* mutant flies have a decreased response to the bitter compound caffeine. Interestingly, ectopic expression of *Ih* in bGRNs also increases sGRN response to sucrose, suggesting that *Ih*-dependent increase in sensillum potential is not specific to *Ih* expressed in sGRNs. The authors further demonstrated, using both SSR and behavior assays, that exposure to sugars in the food substrate is important for the *Ih*-dependent sensitization of bGRNs. The experiments conducted in this paper are of interest to the chemosensory field. The observation that *Ih* is important for the activity in bGRNs albeit expressed in sGRNs is especially fascinating and highlights the importance of non-synaptic interactions in the taste system.

Comments on the revised version:

The authors performed additional analyses/experiments to address my previous major points. I'm satisfied with most of their answers:

- (1) Sensilla types are labeled in all figures. Proper GAL4 and UAS controls were added to the figures.
- (2) Fig. 2A was added to illustrate the important concepts of SP. Fig. 5E was added to show a working model, which could be better but is alright.
- (3) Although not in my list of major points, I appreciate the newly added Fig. 5A and 5B, which demonstrate the long-lasting effect of exposure to sugars.
- (4) Post-stimulus histogram was added for Fig. 4.
- (5) Regarding the expression of *Ih* in bGRNs and sGRNs, the authors referred to their preprint (Lee et al., 2023, Fig 5C, D, suppl movie 1 and 2). The authors stated that "On the other hand, bGRNs labeled by Gr66a-LexA appeared to colocalize only partially with GFP when the confocal stacks were examined image by image." This interpretation unfortunately does not align with my viewing of the images and the movies. Just looking at the images and the movies alone, one would conclude that *Ih* is indeed expressed in both bGRNs and sGRNs. Notably, the *Ih*-TG4.0 is expressed in other non-neuronal cells in the labellum. That being said, I agree with the authors that even if *Ih* is indeed expressed in bGRNs, it would not affect

SP (Fig. 1C, D of this paper, Fig. 5B of Lee et al., 2023 preprint), so I think the authors have addressed my major concern.

<https://doi.org/10.7554/eLife.96602.2.sa1>

### Author response:

The following is the authors' response to the original reviews.

#### **Reviewer #1 (Public Review):**

##### *Summary:*

*This study identifies new types of interactions between Drosophila gustatory receptor neurons (GRNs) and shows that these interactions influence sensory responses and behavior. The authors find that HCN, a hyperpolarization-activated cation channel, suppresses the activity of GRNs in which it is expressed, preventing those GRNs from depleting the sensillum potential, and thereby promoting the activity of neighboring GRNs in the same sensilla. HCN is expressed in sugar GRNs, so HCN dampens the excitation of sugar GRNs and promotes the excitation of bitter GRNs. Impairing HCN expression in sugar GRNs depletes the sensillum potential and decreases bitter responses, especially when flies are fed on a sugar-rich diet, and this leads to decreased bitter aversion in a feeding assay. The authors' conclusions are supported by genetic manipulations, electrophysiological recordings, and behavioral assays.*

##### *Strengths:*

*(1) Non-synaptic interactions between neurons that share an extracellular environment (sometimes called "ephaptic" interactions) have not been well-studied, and certainly not in the insect taste system. A major strength of this study is the new insight it provides into how these interactions can impact sensory coding and behavior.*

We appreciate the reviewer's view that our findings may allow researchers to better understand sensory coding and behavior. However, we respectfully disagree that the SP homeostasis in Drosophila gustation we describe here pertains to ephaptic interaction. Although SP reduction was proposed as the basis of post-ephaptic hyperpolarization in Drosophila olfaction, we find that SP changes are found to be too slow to mediate the fast action of ephaptic inhibition in gustation, reported in the ref#17. We observed a slow, sweet-dependent SP depletion (Fig. 5B, revised), which takes more than one hour. The real-time change of SP was also slow even upon contact with 200-mM sucrose; this result was set aside for another manuscript in preparation. Therefore, we believe the main findings in this paper concern the homeostatic preservation of SP for the maintenance of gustatory function, not ephaptic interaction.

*(2) The authors use many different types of genetic manipulations to dissect the role of HCN in GRN function, including mutants, RNAi, overexpression, ectopic expression, and neuronal silencing. Their results convincingly show that HCN impacts the sensillum potential and has both cell-autonomous and nonautonomous effects that go in opposite directions. There are a couple of conflicting or counterintuitive results, but the authors discuss potential explanations.*

*(3) Experiments comparing flies raised on different food sources suggest an explanation for why the system may have evolved the way that it did: when flies live in a sugar-rich environment, their bitter sensitivity decreases, and HCN expression in sugar GRNs helps to counteract this decrease.*

#### Weaknesses/Limitations:

*(1) The genetic manipulations were constitutive (e.g. Ih mutations, RNAi, or misexpression), and depleting Ih from birth could lead to compensatory effects that change the function of the neurons or sensillum. Using tools to temporally control Ih expression could help to confirm the results of this study.*

We attempted to address this point by using the tub-Gal80ts system. The result is now included as Fig. 1-figure supplement 2. At 29°C, a non-permissive temperature for GAL80ts which allows GAL4-dependent expression Ih-RNAi, we observed that bGRN responses were decreased and sGRN responses were increased compared to the control maintained at 18°C, and this is in parallel with the result in Fig. 1C,D. For this experiment, we inserted “To exclude the possibility that Ih is required for normal gustatory development, we temporally controlled Ih RNAi knockdown to occur only in adulthood, which produced similar results (Fig. 1-figure supplement 2).” (~line 113).

*(2) The behavioral experiment shows a striking loss of bitter sensitivity, but it was only conducted for one bitter compound at one concentration. It is not clear how general this effect is. The same is true for some of the bitter GRN electrophysiological experiments that only tested one compound and concentration.*

We conducted additional behavioral experiments with other bitters such as lobeline and theophylline (Fig. 5-figure supplement 1), which showed sensitivity losses in Ih mutants similar to caffeine. For these results, the following is inserted at ~line 274: “These results were recapitulated with other bitters, lobeline and theophylline (Fig. 5-figure supplement 1).”

We also added single sensillum recording data with bitters, berberine, lobeline, theophylline and umbelliferone, which yielded results similar to those obtained with caffeine (Fig. 1-figure supplement 1). This is described with the sentence at ~line 105 “Other bitter chemical compounds, berberine, lobeline, theophylline, and umbelliferone, also required Ih for normal bGRN responses (Fig. 1-figure supplement 1).”

*(3) Several experiments using the Gal4/UAS system only show the Gal4/+ control and not the UAS/+ control (or occasionally neither control). Since some of the measurements in control flies seem to vary (e.g., spiking rate), it is important to compare the experimental flies to both controls to ensure that any observed effects are in fact due to the transgene expression.*

We appreciate the reviewers for raising this point. Indeed, there was a small logical flaw with the controls. We have now included all the necessary controls for Fig. 1C-F, Fig. 2I,J, Fig. 4E, and Fig. 5D, as reviewers suggested. These experiments remained statistically significant after including the new control groups.

*(4) I was surprised that manipulations of sugar GRNs (e.g. Ih knockdown, Gr64a-f deletion, or Kir silencing) can impact the sensillum potential and bitter GRN responses even in experiments where no sugar was presented.*

We are afraid there is a misunderstanding on the early part of the paper. We suspected that the manipulations impacted bGRNs and SP due to the sweetness in the regular cornmeal food, as stated in lines 214-220 “Typically, we performed extracellular recordings on flies 4-5 days after eclosion, during which they were kept in a vial with fresh regular cornmeal food containing ~400 mM D-glucose. The presence of sweetness in the food would impose long-term stimulation of sGRNs, potentially requiring the delimitation of sGRN excitability for the homeostatic maintenance of gustatory functions. To investigate this possibility, we fed WT

and Ihf03355 flies overnight with either non-sweet sorbitol alone (200 mM) or a sweet mixture of sorbitol (200 mM) + sucrose (100 mM).”

*I believe the authors are suggesting that the effects of sugar GRN activity (e.g., from consuming sugar in the fly food prior to the experiment) can have long-lasting effects, but it wasn't entirely clear if this is their primary explanation or on what timescale those long-lasting effects would occur. How much / how long of a sugar exposure do the flies need for these effects to be triggered, and how long do those effects last once sugar is removed?*

We attempted to address this point with additional experiments (Fig. 5A,B). The reduction of SP could be observed in WT and HCN-deficient mutants with similar degrees 1 hr after the flies were transferred from nonsweet sorbitol-containing vials to sweet sucrose-containing ones. Moreover, the mutants, but not WT, showed further depression of SP when the sweetness persisted in the media for 4 hrs and overnight. This long-term exposure to sweetness longer than 1 hr may simulate the feeding on the regular sweet cornmeal food. The recovery of SP was also tested by removing flies from the sweet media after overnight-long sweet exposure and placing them in sorbitol food. SPs of WT and the mutants were recovered to the similar levels 1 hr after separating the animals from sweetness, although the HCN-lacking mutants showed much lower SP right after overnight sweetness exposure. The unimpaired recovery of the mutants suggests that HCN is independent of generating transepithelial potential itself. Therefore, regardless of HCN, SP changes are not fast even in the presence of strong sweetness, and SP is much better guarded when sGRNs express HCN in a sweet environment.

We inserted the following at ~line 260 to describe the newly added recovery experiment: “Following overnight sweet exposure, SPs of WT and Ihf03355 were recovered to similar levels after 1-hr incubation with sorbitol only food. However, it was after 4 hrs on the sorbitol food that the two lines exhibited SP levels similar to those achieved by overnight incubation with sorbitol only food (Fig. 5B). These results indicate that SP depletion by sweetness is a slow process, and that the dysregulated reduction and recovery of SPs in Ihf03355 manifest only after long-term conditioning with and without sweetness, respectively.”.

*(5) The authors mention that HCN may impact the resting potential in addition to changing the excitability of the cell through various mechanisms. It would be informative to record the resting potential and other neuronal properties, but this is very difficult for GRNs, so the current study is not able to determine exactly how HCN affects GRN activity.*

On this point, we cannot but rely on previous studies of biophysical and electrophysiological characterization on mammalian HCN channels and a heterologous expression study that revealed a robust hyperpolarization-activated cation current from *Drosophila* HCN channels (PMID: 15804582).

#### **Reviewer #2 (Public Review):**

##### *Summary:*

*In this manuscript, the authors start by showing that HCN loss-of-function mutation causes a decrease in spiking in bitter GRNs (bGRN) while leaving sweet GRN (sGRN) response in the same sensillum intact. They show that a perturbation of HCN channels in sweet-sensing neurons causes a similar decrease while increasing the response of sugar neurons. They were also able to rescue the response by exogenous expression. Ectopic expression of HCN in bitter neurons had no effect. Next, they measure the sensillum potential and find that sensillum potential is also affected by HCN channel perturbation. These findings lead them to speculate that HCN in sGRN increases sGRN spiking which in*



turn affects bGRNs. To test this idea that carried out multiple perturbations aimed at decreasing sGRN activity. They found that decreasing sGRN activity by either using receptor mutant or by expressing Kir (a K<sup>+</sup> channel) in sGRN increased bGRN responses. These responses also increase the sensillum potential. Finally, they show that these changes are behaviorally relevant as conditions that increase sGRN activity decrease avoidance of bitter substances.

**Strengths:**

There is solid evidence that perturbation of sweet GRNs affects bitter GRN in the same sensillum. The measurement of transsynaptic potential and how it changes is also interesting and supports the authors' conclusion.

**Weaknesses:**

The ionic basis of how perturbation in GRN affects the transepithelial potential which in turn affects the second neuron is not clear.

We speculate that HCN-dependent membrane potential regulation, rather than ionic composition change, is responsible for the observed SP preservation, as further discussed as an author response in the section of “Recommendations for the authors”. The transepithelial potential can be dissipated by increased conductance through receptor-linked ion channels following gustatory receptor activation in GRNs. The volume of the sensillum lymph is very small according to electron micrographs of horizontally sliced bristles (PMID: 11456419). Therefore, robust excitation of a gustatory neuron may easily deplete the extracellular potential built as a form of polarized ion concentrations across the tight junction. When the consumption is too strong and extended, the neighboring neuron, which share TEP with the activated GRN, can be negatively affected. We propose that HCN suppresses overexcitation of sGRNs by means of membrane potential stabilization. This stabilization prevents sGRNs from excessively reducing the TEP, thereby protecting the activity of neighboring bGRNs.

**Reviewer #3 (Public Review):**

Ephaptic inhibition between neurons housed in the same sensilla has been long discovered in flies, but the molecular basis underlying this inhibition is underexplored. Specifically, it remains poorly understood which receptors or channels are important for maintaining the transepithelial potential between the sensillum lymph and the hemolymph (known as the sensillum potential), and how this affects the excitability of neurons housed in the same sensilla.

Although a reduction of sensillum potential was proposed to underlie membrane hyperpolarization of post-ephaptic olfactory neurons in *Drosophila*, our preliminary data (not shown due to a manuscript in preparation) and the results included in the paper (Fig. 5B) strongly suggest that SP reduction is not a requisite for ephaptic inhibition at least in GRNs. Ephaptic inhibition is expected to be instantaneous, whereas we find that SP reduction in gustation is very slow. Therefore, we would like to indicate that the findings we report in this manuscript are not directly related to ephaptic inhibition.

Lee et al. used single-sensillum recordings (SSR) of the labellar taste sensilla to demonstrate that the HCN channel, *I<sub>h</sub>*, is critical for maintaining sensillum potential in flies. *I<sub>h</sub>* is expressed in sugar-sensing GRNs (sGRNs) but affects the excitability of both the sGRNs and the bitter-sensing GRNs (bGRNs) in the same sensilla. *I<sub>h</sub>* mutant flies have decreased sensillum potential, and bGRNs of *I<sub>h</sub>* mutant flies have a decreased response to the bitter compound caffeine. Interestingly, ectopic expression of *I<sub>h</sub>* in bGRNs also increases sGRN response to sucrose, suggesting that *I<sub>h</sub>*-dependent increase in sensillum potential is not specific to *I<sub>h</sub>* expressed in sGRNs. The authors further demonstrated,

*using both SSR and behavior assays, that exposure to sugars in the food substrate is important for the Ih-dependent sensitization of bGRNs. The experiments conducted in this paper are of interest to the chemosensory field. The observation that Ih is important for the activity in bGRNs albeit expressed in sGRNs is especially fascinating and highlights the importance of non-synaptic interactions in the taste system.*

*Despite the interesting results, this paper is not written in a clear and easily understandable manner. It uses poorly defined terms without much elaboration, contains sentences that are borderline unreadable even for those in the narrower chemosensory field, and many figures can clearly benefit from more labeling and explanation. It certainly needs a bit of work.*

We would like to revise the language aspect of the manuscript after finalizing the scientific revision.

*Below are the major points:*

*(1) Throughout the paper, it is assumed that Ih channels are expressed in sugar-sensing GRNs but not bitter-sensing GRNs. However, both this paper and citation #17, another paper from the same lab, contain only circumstantial evidence for the expression of Ih channels in sGRNs. A simple co-expression analysis, using the Ih-T2A-GAL4 line and Gr5a-LexA/Gr66a-LexA line, all of which are available, could easily demonstrate the co-expression. Including such a figure would significantly strengthen the conclusion of this paper.*

We did conduct confocal imaging with Ih-T2A-Gal4 in combination with GRN Gal4s (ref#17 version2). The expression is very broad, including both neurons and non-neuronal cells. We observed much stronger sGRN expression than bGRN expression. But the promiscuous expression of the reporter in many cells hindered us from clearly demonstrating the void of the reporter in bGRNs. However, the functional and physiological examination of Ih-T2A-Gal4 with the neuronal modifiers such as TRPA1 and Kir2.1 in ref#17 indicates the strong and little expression of Ih in sGRNs and bGRNs, respectively. Furthermore, the RNAi kd results present another line of evidence that HCN expressed in sGRNs regulates SP and bGRN activity (Fig. 1C,D, Fig. 1-figure supplement 2). Ih-RNAi expression in bGRNs did not result in any statistically significant changes in the activities of sGRNs and bGRNs compared to controls (Fig. 1C,D, revised), advocating that Ih acts in sGRNs for the functional homeostasis of SP and GRNs, as we claim.

*(2) Throughout this paper, it is often unclear which class of labellar taste sensilla is being recorded. S-a, S-b, I-a, and I-b sensilla all have different sensitivities to bitters and sugars. Each figure should clearly indicate which sensilla is being recorded. Justification should be provided if recordings from different classes of sensilla are being pooled together for statistics.*

We mainly performed SSR (single sensillum recording) on i-type bristles as they have the simplest composition of GRNs compared to s- and L-type bristles. As single s-types also contain each of s- and bGRN, we measured SP also for s-types (Figs. 2, 3F and 4D). In case of Fig.3-figure supplement 1, L-types were tested for the relationship between water cell activity and SP. Now all the panels are labelled with the tested bristle types.

*(3) In many figures, there is a lack of critical control experiments. Examples include Figures 1C-F (lacking UAS control), Figure 2I-J (lacking UAS control), Figure 4E (lacking the UAS and GAL4 control, and it is also strange to compare Gr64f > RNAi with Gr66a > RNAi, instead of with parental GAL4 and UAS controls.), and Figure 5D (lacking UAS control).*

*Without these critical control experiments, it is difficult to evaluate the quality of the work.*

Thank you for pointing this out. We appreciate the feedback and have addressed these concerns by including all the requested controls in the figures. Specifically, we have added the UAS controls for Figs 1C-F and 2I-J, as well as the UAS and GAL4 controls for Fig. 4E. We have also included the UAS control for Fig. 5D.

*(4) Figure 2A could benefit from more clarification about what exactly is being recorded here. The text is confusing: a considerable amount of text is spent on explaining the technical details of how SP is recorded, but very little text about what SP represents, which is critical for the readers. The authors should clarify in the text that SP is measuring the potential between the sensillar lymph, where the dendrites of GRNs are immersed, and the hemolymph. Adding a schematic figure to show that SP represents the potential between the sensillar lymph and hemolymph would be beneficial.*

SP was defined at lines 55-56 in the first paragraph of introduction, which also contains the background information for SP as a transepithelial potential. As reviewer suggested, we now also included a sentence describing SP (“SP is known as a transepithelial potential between the sensillum lymph and the hemolymph, generated by active ion transport through support cells”, line 126) and a drawing to illustrate the concept of SP (Fig. 2A), and revised the legend.

*(5) The sGRN spiking rate in Figure 4B deviates significantly from previous literature (Wang, Carlson, eLife 2022; Jiao, Montell PNAS 2007, as examples), and the response to sucrose in the control flies is not dosage-dependent, which raises questions about the quality of the data. Why are the responses to sucrose not dosage-dependent? The responses are clearly not saturated at these (10 mM to 100 mM) concentrations.*

Our recordings show different spiking frequencies from others’ work, because the frequencies are from 5-sec bins not only first 0.5 sec. This lowers the frequencies, as spikes are relatively more frequent in the beginning of the recording (Fig. 4-figure supplement 1).

*Why are the responses to sucrose not dosage-dependent? The responses are clearly not saturated at these (10 mM to 100 mM) concentrations.*

We were also puzzled with the flat dose dependence to sucrose. This result may suggest the existence of another mechanism moderating sucrose responses of sGRNs. This flat curve reappeared with other genotypes with the same concentration range (5-50 mM) in Fig. 4E. However, 1-mM sucrose produced much lower spiking frequencies (Fig. 4E), suggesting that sGRN responses are saturated at 5 mM sucrose with our recording/analysis condition.

*(6) In Figure 4C, instead of showing the average spike rate of the first five seconds and the next 5 seconds, why not show a peristimulus time histogram? It would help the readers tremendously, and it would also show how quickly the spike rate adapts to overexpression and control flies. Also, since taste responses adapt rather quickly, a 500 ms or 1 s bin would be more appropriate than a 5-second bin.*

Taste single sensillum recording starts by contacting stimulants, which bars us from recording pre-stimulus responses of GRNs. Therefore, we showed post-stimulus graphs with 1-sec bins (Fig. 4-figure supplement 1) as we reviewer suggested.

*(7) Lines 215 - 220. The authors state that the presence of sugars in the culture media would expose the GRNs to sugar constantly, without providing much evidence. What is*

*the evidence that the GRNs are being activated constantly in flies raised with culture media containing sugars? The sensilla are not always in contact with the food.*

We agree with reviewer. We replaced “long-term stimulation of sGRNs” with “strong and frequent stimulation of sGRNs for extended period”. The word long-term may be interpreted to be constant.

*(8) Line 223. To show that bGRN spike rates in Ih mutant flies "decreased even more than WT", you need to compare the difference in spike rates between the sorbitol group and the sorbitol + sucrose group, which is not what is currently shown.*

The data were examined by ANOVA and a multiple comparison test (Dunn's) between all the groups regardless of genotypes and conditions in the panel (all the groups sharing the y axis). Therefore, the differences were statistically examined. However, the cited expression we used read like it was about the slope or extent of the decrease. We intended to indicate the difference in the absolute values of spiking frequencies after overnight sweet exposure between the genotypes, while bGRN activities were statistically indifferent between WT and Ih mutants when they were kept only on sorbitol food. We revised it to “decreased to the level significantly lower than WT”. We also changed the graph style to effectively present the trend of changes in bGRN sensitivity with comparison between genotypes. Again, the groups were statistically examined together regardless of the genotypes and conditions.

*(9) To help readers better understand the proposed mechanisms here, including a schematic figure would be helpful. This should show where Ih is expressed, how Ih in sGRNs impacts the sensillum potential, how elevated sensillum potential increases the electrical driving force for the receptor current, and affects the excitability of the bGRNs in the same sensilla, and how exposure to sugar is proposed to affect ion homeostasis in the sensillum lymph.*

As reviewer suggested, we included two panels to show working model for gustatory homeostasis via SP maintenance by HCN (Fig. 5E,F).

#### **Reviewer #1 (Recommendations For The Authors):**

*(1) The relationship between this paper and the authors' bioRxiv preprint posted last year is not clear. In the introduction they made it seem like this paper is a follow-up that builds on the preprint, but most or all of the experiments in this paper were already performed in the preprint. I guess the authors are planning to divide the original paper into two papers. I would suggest updating the preprint to avoid confusion.*

Thank you for the comment. We updated the preprint to be without a part of Fig.6 and entire Fig.7 along with associated texts. As reviewer pointed out, our eLife paper was spun off from the part of the preprint paper, because we feel that the two stories could confuse readers when presented together.

*(2) Have the authors considered testing responses of water GRNs? They reside in the same sensilla as sugar neurons, so are they also increased affected by Ih mutation or RNAi in sugar neurons? This would strengthen the evidence that the indirect (non-cell autonomous) effects of Ih are due to the sensillum potential and not some specific interaction between sweet and bitter cells.*

As reviewer proposed, we appraised water GRN activity in the L-type bristles of WT, Ihf03355 and a genomic rescue line for Ihf03355. Spiking responses in water GRNs were evoked by hypo-osmolarity of electrolyte (0.1 mM tricholine citrate-TCC). Interestingly, the Ih mutant

showed reduced 0.1 mM TCC-provoked spiking frequencies compared to WT. This impairment was rescued by the genomic fragment containing an intact Ih locus (Figure 3-figure supplement 1A).

Additionally, SPs in L-type bristles were reduced by Ih deficiencies but increased in Gr64af, suggesting that HCN regulates sGRNs in L-type bristles as well (Figure 3-figure supplement 1B). Again, the bristles of animals with both mutations together exhibited SPs similar to those of WT.

Furthermore, when we conducted cDNA rescue experiments in L bristles, introduction of Ih-RF cDNA in sGRNs restored SPs, while expressing it in bGRNs did not unlike the results from the i- and s-bristles (Fig. 2K,L), likely because L-bristles lack bGRNs. These cDNA rescue and genetic interaction experiments were conducted using flies fed on fresh cornmeal food with strong sweetness, suggesting that the sweetness in the media is the likely key factor producing the genetic interaction and necessitating HCN, consistent with other results in the manuscript. Therefore, SP regulation by HCN is observed in the L-type bristles.

*Minor comments:*

*Line 52: typo, "Many of"*

Thank you. Corrected

*Line 95: typo, "sensilla do an sGRN"*

Corrected

*Line 98: typo, "we observed reduced the spiking responses"*

Corrected

*Line 206: typo, "a relatively low sucrose concentrations"*

Corrected

*Line 260: "inverse relationship between the two GRNs in excitability" - I am not exactly sure what data you are referring to.*

Although alleles did not show increased sGRN activities, knockdown of Ih decreased bGRN activity but increased sGRN activity (Fig. 1C,D, Fig.1-figure supplement 2B), while suppression of sGRNs increased bGRN activity (Fig. 3). To clarify this point, we revised the phrase to "the inverse relationship between the two GRNs in excitability observed in Fig. 1C,D, Fig. 1-figure supplement 2B, and Fig. 3".

*Methods: typo, "twenty of 3-5 days with 10 males and 10 females"*

Corrected to "Twenty flies, aged 3-5 days and consisting of 10 males and 10 females,"

*Methods: typo, "Kim's wipes" should be "Kimwipes"*

Corrected

**Reviewer #2 (Recommendations For The Authors):**

*(1) More clarification is necessary on Transepithelial potential (TEP). TEP is typically created by having pumps and tight junctions between the sensillar lymph and the hemolymph.*

We have an introduction to TEP or SP in the context of sensory functions (lines 40-57) with relevant references. The involvement of pumps and tight junction was mentioned in the same paragraph; “Glia-like support cells exhibit close physical association with sensory receptor neurons, and conduct active transcellular ion transport, which is important for the operation of sensory systems” (line 40) and “Tight junctions between support cells separate the externally facing sensillar lymph from the internal body fluid known as hemolymph” (line 53).

*It is not clear how HCN channels in one of the neurons might change the composition of the sensillum lymph. An explanation of their model of how TEP depends on HCN is necessary.*

Although the ionic composition of the sensillum lymph is a contributing factor to the sensillum potential, it is more conceptually relevant to describe our findings with the perspective of membrane potential regulation given the role of HCN in membrane potential stabilization as discussed in our manuscript.

We speculate that HCN controls the membrane potential at rest and/or in motion to modulate sGRN activity towards saving SP despite the sweetness in the niche. We positioned our results in relation to SP in discussion; “Our results provide multiple lines of evidence that HCN suppresses HCN-expressing GRNs, thereby sustaining the activity of neighboring GRNs within the same sensilla. We propose that this modulation occurs by restricting SP consumption through HCN-dependent neuronal suppression rather than via chemical and electrical synaptic transmission.” (lines 252-255). Moreover, it is unclear whether HCN is localized to the dendrite bathed in the sensillum lymph to influence the ionic composition of the lymph. It would be very interesting to study in future whether the ionic flow through HCN channels itself is critical for the function of HCN in this context, and whether HCN is exclusively present in the dendrite to support the postulation. However, we would like to remind reviewer that Kir2.1 and HCN channels in sGRNs showed similar effects on SP and bGRNs, while they differ in Na<sup>+</sup> conductance.

In the initially submitted manuscript (lines 325-343), we discussed the potential mechanism by which Kir2.1 and HCN channels commonly increase SP in terms of how the membrane potential regulation in the soma can control the SP consumption in the dendrite of sGRNs.

*Another point about the TEP that needs some explanation is that these sensilla are open to the environment as tastants must flow in and are different from mechanical sensilla in that sense.*

This is a very important question regarding the general physiology of the taste sensilla, as the sensillum lymph is in contact with the external environment through the pore of the sensillum. It is indeed interesting to consider how the composition and potential of the lymph are maintained despite the relatively vast volume of food the sensilla encounter during gustation and the continuous evaporation to air between episodes of gustation. However, we believe that this question, while important, is distinct from the primary focus of our manuscript.



*Are the TEP measurements in Figure 2 under control conditions where there are no tastants?*

There is no tastant in the SP-measuring glass electrode other than the electrolyte. We apologize that we did not specify the recording electrode condition. We inserted a clause in the method; “For SP recordings, the recording electrode contained 2 mM TCC as the electrolyte, and...”

*Does the TEP change dynamically as sGRN is activated?*

SP does shift in response to sweets. Please see Fig. 5B. Also, we showed SP changes by mechanical stimuli, which depended on the mechanoreceptor, NompC (Fig. 2D-F). Mechanoreceptor neurons share the sensillum lymph with GRNs.

*(2) More clarification on the potential transduction mechanism and how TEP affects one neuron differentially. Essentially, sGRN perturbation affects sGRN activity and it affects the TEP. More explanation is needed for the potential ionic mechanism of each.*

Our results strongly suggest that HCN lowers the activity of HCN-expressing GRNs, mitigating SP consumption. This modulation is crucial because the SP serves as a driving force for neuronal activation within the sensillum. HCN is particularly necessary in sGRNs because of the flies’ sweet feeding niche, which is expected to result in frequent and strong activation of sGRNs. The SP saved by HCN-dependent delimitation of sGRNs can be used to raise the responsibility of bGRNs.

*(3) The authors refer to their own unreviewed paper (Reference 17). This paper is on a similar topic and there seems to be some overlap. Clarification on this point would be important.*

We revised the biorxiv preprint, so that the preprint version 2 does not contain the parts overlapping with this eLife paper. This eLife paper was originally part of the preprint paper, but it was separated to clarify the messages of the two stories. As we explained in Discussion (lines 276-297), HCN provides resistance to both hyperpolarization and depolarization of the membrane potential. Simply put, one paper focuses on the role of HCN in resisting hyperpolarization, while the other (this paper in eLife) focuses on resisting depolarization.

*(4) Methods are sparse. Many details on the method are necessary. For example, Sensilla recordings are being done by the tip-dip method (I assume). What does "number of experiments" mean in Figure 1? Is it the number of animals or the number of sensilla? How many trials/sensilla?*

We indicated the extracellular recording was performed by the tip-dip method; “In vivo extracellular recordings were performed by the tip-dip method as detailed previously”. We also added a statement on the number of experiments; “The number of experiments indicated in figures are the number of naïve bristles tested. The naïve bristles were from at least three different animals.”

*(5) Figure 1: I understand the author's interpretation. But if one compares WT in Figure 1A to Gr64a-IhRNAi in 1C, we can come to the conclusion that there is no change. In other words, the control in Figure 1C (grey) has a much higher response than WT. Similar conclusions can be made for other experiments. Is the WT response stable enough to make the conclusions made here?*

The genetic background of each genotype may influence GRN activity to some extent. RNAi knockdown experiments are well-known for their hypomorphic nature, and their effects should be evaluated by comparison with their parental controls such as Gal4 and UAS lines. As all reviewers pointed out, we added the results from UAS control. This effort confirms that Gr89a>Ih RNAi is statistically indifferent to UAS control as well as Gr64f-Gal4 control in bGRN spiking evoked by 2-mM caffeine, while Gr64f>Ih RNAi showed reduced bGRN responses to 2 mM caffeine compared to all the controls.

(6) *Figure 3: Why is bGRN spiking not plotted against sensillum potential to observe the dependence more directly?*

This is a very interesting suggestion. We are not, however, equipped to measure spiking and sensillum potential simultaneously. Therefore, they are independent experiments, and we treated them accordingly.

(7) *Figure 4: Why bGRN response is only affected at high caffeine concentrations is not clear.*

We were also surprised by the differences in the dose dependence results of b- and sGRNs, genetically manipulated to mis-express and over-express HCN in Fig. 4A and 4E, respectively. Each gustatory neuron likely has distinct sets of players and parameters that set its own membrane potential and excitability.

We can think of a possibility that there might be a range of membrane potentials within which HCN does not engage. In bGRNs, the resting membrane potential may lie low within this range, so that some degrees of membrane depolarization by low concentrations of caffeine do not significantly close HCN channels, thus preventing their hyperpolarizing effects. On the other hand, the membrane potential of sGRNs may be high within this range, showing suppressive effects at all tested sucrose concentrations. However, we find this explanation is too speculative to include in the main text, while we stated in the original manuscript, “implying a complex cell-specific regulation of GRN excitability.” (line 210).

(8) *Minor:*

*L98 - there is a small typo*

Corrected

*L274: "funny" !?*

“Funny” currents, denoted If, were initially observed by electrophysiologists and later attributed to HCN channels, now indicated by Ih (thus the gene name Ih in *Drosophila*). These currents were termed “funny” due to their unusual properties compared to other currents. For more detailed information, please refer to the cited references.

*L257: Neuropeptide seemed to be abrupt*

We attempted to discuss possible mechanisms that mediate excitability changes across GRNs beyond the mechanism by SP shifts. Neuropeptides, which are chemical neurotransmitters along with small neurotransmitters, were mentioned following the discussion on synaptic transmission to suggest alternative pathways for excitability regulation. This inclusion is meant to provide a comprehensive overview of potential mechanisms influencing GRN activity.

**Reviewer #3 (Recommendations For The Authors):**

*Congratulations on your fascinating research! The results are certainly of interest to the chemosensory field. However, I suggest using academic editing services to enhance the clarity of your text and ensure that the terminology and jargon align with standard usage in the field. The current choice of words may not be consistent with commonly used terms. As it is now, the writing might not fully showcase the compelling story and the effort behind your study, and is underselling your interesting results. Proper refinement could make sure your valuable findings are appropriately recognized.*

We appreciate your comments and apologize for any difficulties reviewers faced during the review process. We are currently prioritizing the review of scientific content and plan to address language issues in a subsequent revision. It would be very helpful for future revisions if the problematic sentences or expressions could be indicated in detail after this revision. This will allow us to ensure that our terminology and expression align with standard usage in the field, and that our findings are clearly and effectively communicated.

**Minor points:**

(1) Line 110: what is Ih-RF?

We apologize that we relied on a reference in describing the cDNA. The following clause was inserted with additional reference and the Flybase id: “(Flybase id: FBtr0290109), which previously rescued Ih deficiency in other contexts<sup>17,26</sup>,”

(2) Line 158: *Gr64af* mutant flies still have *Gr5a* and a residual response to fructose and sucrose (Slone, Amrein 2007).

We revised the line to “is severely impaired in sucrose and glucose sensing”, since there is a substantial loss of sucrose and glucose sensing in both *Gr64af* from Kim et al 2018 and *DGr64* from Slone et al 2007, when they were examined by the proboscis extension reflex assay. This was also confirmed in the study by Jiao et al 2009. We also deleted “sugar-ageusic” and instead describe the mutant “impaired in sucrose and glucose sensing” in Fig. 3 legend.

(3) Lines 264-273 seem unnecessary. This paper is not about the function of HCN in mammals, and these discussions seem largely irrelevant.

We feel that it is important to position our results within a broader context by discussing the potential implications of our findings for sensory systems of other animals. As we stated, HCN channels have been localized in mammalian sensory systems, but their roles are often not well understood. By including this discussion, we aim to highlight the relevance of our findings beyond the model organism used in our study and suggest possible areas for future research in mammalian systems.

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